SIGNALS AND SYSTEMS

Spatially-Weighted Chrominance Fusion

SW-CF / Contactless pulse estimation from RGB video

An implemented Python research prototype with five-region quality weighting, POS projection, digital filtering, and spectral pulse-rate estimation.

Evidence status
Software tests and synthetic validation are included. Real webcam-reference and UBFC accuracy have not been measured. Pulse-interval variability is experimental; no validated stress score is reported.

Team, roll numbers, institution, and supervisor:
Add your own details in docs/REPORT.md before submission.

Submission edition: 27 October 2026

Abstract

This project implements a software prototype for extracting a contactless pulse signal from an ordinary RGB webcam. Five facial regions are located with MediaPipe landmarks. Their spatial mean colors form sampled RGB time series. Brightness, clipping, and landmark motion provide physical quality gates; a spectral concentration score supplies temporal quality information. A sliding Plane-Orthogonal-to-Skin (POS) projection extracts a pulse candidate from each region, followed by adaptive weighted fusion, Butterworth bandpass filtering, and Welch power spectral density estimation. The interface displays region weights, a normalized waveform, an estimated pulse rate, and an experimental pulse-interval variability statistic. Tests and artificial color signals verify the software and expected DSP behavior. An evaluator is included for future UBFC-rPPG validation. No clinical accuracy or stress-detection performance is claimed.

1. Problem statement and objectives

Contact sensors and wearables are unnecessary for a classroom demonstration of remote photoplethysmography (rPPG). Changes in reflected skin light can contain a weak blood-volume-related periodic component. Camera motion, illumination, compression, and automatic exposure often dominate that component. A single poor facial region can therefore corrupt an otherwise useful measurement.

The engineering objective is to construct and demonstrate a reproducible pipeline that combines several regions according to their quality, rather than depending on a single patch. The system must expose its intermediate behavior, reject obviously inadequate input, export numeric signals, and support comparison with simpler baselines. A further objective is to explain the sampling, filtering, estimation, and latency tradeoffs using Signals and Systems concepts.

The deliverable is an educational research prototype. It does not diagnose conditions, classify stress, or replace a pulse oximeter or ECG. POS and rPPG are established methods; SW-CF is this project's integration and weighting design, not a claim of a newly invented physiological technique [1].

2. System architecture

```
RGB camera / local video
          |
 timestamped frame acquisition
          |
 MediaPipe face landmarks
          |
 five interior facial polygons
          |
 mean RGB + light/clipping/motion quality
          |
 timestamp resampling -> per-region POS -> bandpass
          |
 spectral SQI -> normalized, gated fusion -> bandpass
          |
 Welch PSD -> BPM / rejection -> dashboard and CSV
          |
 separate long history -> screened pulse intervals -> SDNN proxy
```

The regions are forehead, left cheek, right cheek, nose, and chin. Compact polygons use indices shared with the original 468-point face mesh. The current MediaPipe Tasks model supplies additional points; unused points are ignored. Polygons shrink toward their centers to reduce boundary contamination. This is geometric region selection, not a learned pixel-level skin classifier [2].

3. Acquisition, sampling, and buffering

The application requests 640x480 or 1280x720 frames at 30 FPS. On Windows it prefers DirectShow; on Linux it prefers V4L2. A driver buffer size of 1 is requested, and a capture thread maintains a single latest-frame slot. This limits application backlog when computation falls behind, but camera exposure, driver queues, and operating-system scheduling remain outside the application's control. OpenCV explicitly documents backend-dependent capture properties [3].

Live timestamps come from a monotonic high-resolution clock at frame delivery. They are delivery-time proxies, not sensor exposure timestamps. Recorded videos use frame index divided by reported FPS, assuming constant-frame-rate media. Consequently processing a video faster than real time does not alter its pulse frequency.

The short analysis window is six seconds: approximately 180 frames at 30 FPS. A time-based deque also retains up to 65 seconds for variability analysis. Frames are linearly resampled onto a uniform grid before digital filtering. The effective sample rate is (N - 1)/(last timestamp - first timestamp). Capture gaps over 0.25 seconds trigger reacquisition. Analysis requires an effective sample rate of at least 8 Hz. Neither interpolation nor frame discarding can recover information lost to acquisition.

4. Spatial signals and quality scores

For patch k and color channel c, the spatial measurement is:

```
C[k,c,n] = mean of channel c over pixels inside patch k at frame n
```

The per-frame physical quality q[k,n] lies in [0,1]. A soft brightness score penalizes very dark or very bright regions; the fraction of near-black or near-saturated pixels reduces quality further. Landmark displacement is normalized by face width and elapsed time to express motion in face-widths per second. A completely off-frame patch receives zero quality.

```
q = light_score * (1 - clipped_fraction)^2 * exp(-3 * normalized_motion)
```

These are transparent engineering heuristics, not a learned confidence probability or a skin-tone-calibrated model. The mean physical score is combined with spectral concentration to form patch SQI. If a patch is contaminated by a periodic lighting artifact, it may still look spectrally strong. This limitation cannot be resolved by SQI alone.

5. POS color-space projection

Wang et al. proposed POS to reduce intensity-related interference by projecting temporally normalized RGB signals onto a plane orthogonal to a normalized skin direction [1]. The implementation uses sliding 1.6-second windows:

```
c(t)  = RGB(t) / temporal_mean(RGB) - 1
s1(t) = g(t) - b(t)
s2(t) = g(t) + b(t) - 2*r(t)
alpha = std(s1) / max(std(s2), epsilon)
h(t)  = s1(t) + alpha*s2(t)
```

Each local h is mean-centered and overlap-added. Counts normalize how many windows contributed to each sample. POS is applied separately to each patch so that regions with different DC skin appearance and lighting are normalized locally. Vectorized sliding-window calculations keep the long-history analysis practical on a CPU. This processing is time varying and data dependent, rather than one fixed linear time-invariant operation.

6. Digital filtering and weighted fusion

The passband is 0.75-2.5 Hz, corresponding to 45-150 BPM. Linear detrending precedes filtering. SciPy second-order sections provide better numerical conditioning than a high-order direct-form polynomial implementation [4].

The code uses `butter(4, [0.75, 2.5], btype='bandpass', fs=fs, output='sos')`. Here 4 is the low-pass prototype order: the resulting bandpass has eight poles. Forward/backward `sosfiltfilt` processing has zero phase on the buffered signal and squares the one-pass magnitude response. Therefore nominal one-pass cutoff points have about -6 dB combined gain, rather than -3 dB. The project describes this accurately instead of calling the whole operation simply fourth order.

For each region, spectral concentration rho is the ratio of in-band PSD within 0.18 Hz of its dominant frequency to total passband PSD. Half a second at each end of the window is excluded from the SQI/PSD calculation to reduce edge effects. Region signals are standardized by their interior standard deviation. Constant regions receive zero score.

```
SQI[k] = mean(q[k,:]) * rho[k]
score[k] = SQI[k]^2
w[k,n] = q[k,n]*score[k] / sum_j(q[j,n]*score[j])
fused[n] = sum_k(w[k,n] * standardized_pulse[k,n])
```

The fused signal is bandpass-filtered again to reduce modulation caused by changing weights. Normalized time-average weights are shown in the interface. Equal-POS and mean-green baselines are included for ablation. They still use the shared acquisition/quality rejection machinery, so they are not completely ungated end-to-end algorithms.

Zero phase is achieved with samples already stored in the window. This is not strictly causal streaming or a promise of zero latency. Newly plotted edge samples can change as later windows arrive. Approximately six seconds of initial acquisition and a half-second update cadence are intentional design tradeoffs.

7. Heart-rate estimation

Welch's method estimates PSD using Hann-windowed segments of up to six seconds. The dominant peak inside the allowed passband determines the pulse-rate estimate [5]. A local parabolic interpolation of log power refines the displayed peak position:

```
estimated_BPM = 60 * estimated_peak_frequency_Hz
```

Zero padding provides a finer numerical frequency grid; it does not improve the information-theoretic resolution of a short observation. A six-second window has a basic frequency scale near 1/6 Hz, or 10 BPM. Trimming and Hann windowing broaden spectral features further. A clean single sinusoid can still yield a small interpolated error, but this does not justify the same accuracy claim for noisy human video.

BPM is withheld when fused physical quality is below 0.20, dominant-band concentration is below 0.40, or the signal has negligible amplitude. Face loss clears the acquisition state. A rejected short window breaks long-history continuity. These gates reject several obvious failures but cannot prove that a periodic signal is physiological or correctly distinguish every harmonic.

8. Experimental pulse-interval variability

A separate history of at least 61 seconds is used before attempting interval statistics. Peaks require minimum spacing and prominence; local parabolic refinement provides a sub-frame timing estimate. The first and last half-second are excluded. Intervals outside 400-1334 ms or more than 20% from the median are rejected. At least 90% must pass, and median interval rate must agree with the spectral estimate. The remaining interval sample standard deviation is reported in milliseconds using N-1 normalization.

```
SD_pulse = sqrt(sum((IBI_i - mean_IBI)^2) / (M - 1))
```

The interface calls this an experimental SDNN proxy. It is pulse-rate variability, not verified ECG normal-to-normal HRV: the application cannot identify ectopic beats reliably. Screening can bias variability downward and reject real variation. Thirty-FPS sampling gives about 33.3 ms between frames; interpolation cannot restore missing beat-timing information. Published HRV guidance emphasizes duration and artifact handling, with conventional short-term comparisons commonly using five-minute recordings [6,7].

SDNN depends on duration, posture, breathing, rhythm, and measurement quality. No validated mapping from this proxy to psychological stress is implemented. The original 'stress monitor' idea is retained as a future research direction, requiring an appropriate study and reference labels.

9. Implementation and reproducibility

The tested environment is Windows, Python 3.11.9, NumPy 2.4.6, SciPy 1.17.1, OpenCV contrib 4.14.0.94, and MediaPipe 0.10.35. The implementation uses the current Tasks face-landmarker interface rather than relying on legacy `mp.solutions.face_mesh`. Python 3.11 is recommended for reproducing the run. Pinned installed versions are supplied in `requirements-lock.txt`.

The application supports webcam, constant-frame-rate local video, simulated RGB input, and UBFC evaluation. It exports raw patch means, quality, accepted and rejected BPM estimates, final pulse waveform, session metadata, and a dashboard image. No webcam video file is recorded. Dependency/model downloads are separate setup actions; image processing occurs on the laptop. Upstream MediaPipe terms and possible package telemetry are distinct from application video handling [8].

The test suite covers six target pulse rates, timestamp jitter and ordering, frame gaps, no-pulse rejection, poor lighting, corrupted regions, filter attenuation, face-loss reset, interval warm-up, UBFC parsing, coverage accounting, and real model initialization on blank frames. The simulation script generates fixed-seed numeric results and figures. An optional public-portrait fixture checks landmark geometry and the full video path with deliberately artificial color modulation. It is not a measurement of the pictured person's pulse.

10. Results and evidence boundaries

The generated synthetic experiment uses target rates of 48, 60, 72, 90, 120, and 144 BPM; three random seeds; clean and corrupted conditions; and three fusion modes. Each condition/mode group contains 18 ten-second artificial signals with timestamp jitter. The corruption affects two regions. Synthetic physical quality labels are supplied by the generator as oracle annotations, so this test isolates fusion and does not validate camera quality estimation.

Condition	Mode	Accepted / cases	MAE (BPM)
Clean synthetic	Weighted POS	18 / 18	0.042
Clean synthetic	Equal POS	18 / 18	0.042
Clean synthetic	Mean green	18 / 18	0.037
Corrupted synthetic	Weighted POS	18 / 18	0.042
Corrupted synthetic	Equal POS	18 / 18	0.252
Corrupted synthetic	Mean green	18 / 18	0.230

These values come from `results/synthetic_results.csv` and `results/synthetic_summary.json`. They are intentionally rounded in this report. The favorable corruption result is conditional on accurate supplied quality labels. It is a software demonstration, not evidence of superior performance on real subjects. Green performs slightly better on this uncomplicated clean simulation; there is no universal superiority claim.

The 70-second dashboard simulation with a 72 BPM target produced a final estimate of approximately 72.14 BPM and an experimental interval SD near 5.27 ms. Its ideal input has constant rate, so nonzero interval SD represents noise/processing error rather than real physiological variability. The color-modulated still-image video yielded approximately 72.08 BPM through the full face-tracker path. Neither is human pulse validation.

The automated test suite passes. Exact test output is included in `results/test_run.txt`. No actual webcam-reference accuracy or real UBFC score has been measured, and no subject-study results are invented.

11. UBFC benchmark protocol

The evaluator targets UBFC-rPPG Dataset 2 folders containing `vid.avi` and `ground_truth.txt`. The authors describe RGB videos recorded at 30 FPS and 640x480, with synchronized CMS50E pulse-oximeter reference data [9]. Users must obtain the dataset from the authors under the applicable terms.

Three ground-truth rows are parsed as PPG waveform, device BPM, and timestamps. The first reference timestamp is subtracted. Video time is frame index/FPS, assuming simultaneous starts and constant frame rate. No lag is optimized. Mean supplied device BPM within each trailing six-second window is compared with the estimate every 0.5 seconds. The measured PPG row is loaded and validated, but is not used to compute the current reference BPM metric.

MAE, RMSE, signed bias, accepted-window coverage, per-subject results, and failed subjects are saved. Missing predictions stay missing. Coverage uses all reference-covered evaluation windows, including face-loss/rejection windows, rather than counting only successful estimates. Reference-uncovered windows are excluded. Correlated overlapping windows must not be treated as independent subjects for statistical significance. No train/test leakage is involved in this fixed-rule prototype, but future threshold tuning should use a separate development set.

12. Limitations and future work

The system assumes one mostly frontal, still face in adequate steady light. Facial hair, makeup, occlusions, low resolution, motion, compression, changes in auto-exposure, rolling shutter, and varying skin appearance can reduce performance. Region quality thresholds are heuristics and have not been calibrated across people or cameras. No fairness or subgroup accuracy claim is made. Switching subjects requires a reset.

All patches can contain the same periodic illumination artifact; a strong PSD peak alone does not prove a heartbeat. The 45-150 BPM band excludes rates outside that range by design. Short windows trade rate stability for responsiveness. Linear timestamp interpolation assumes sufficiently dense samples. The displayed waveform has arbitrary units and cannot be interpreted as blood pressure, oxygen saturation, or ECG morphology.

Future work should evaluate UBFC and additional datasets, quantify errors and coverage by motion/lighting conditions, compare longer windows, test robust skin masks and motion compensation, and measure CPU latency on specified hardware. Experimental variability needs longer recordings and reliable beat-level reference timing. Stress modeling would require validated study labels, subject-separated evaluation, and evidence beyond SDNN alone.

13. Conclusion

SW-CF provides an implemented, inspectable classroom pipeline from RGB video to quality-weighted pulse estimation. It demonstrates spatial averaging, temporal normalization, sampled-data processing, spectral estimation, adaptive fusion, and quality rejection in one application. Synthetic tests verify the software's expected behavior; real-subject accuracy and stress monitoring remain validation tasks. The project is ready for a transparent software demonstration and subsequent dataset experiments.

References

[1] Wang, W., den Brinker, A. C., Stuijk, S., and de Haan, G. (2017). Algorithmic Principles of Remote-PPG. IEEE Transactions on Biomedical Engineering, 64(7), 1479-1491. https://doi.org/10.1109/TBME.2016.2609282

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[7] Shaffer, F., and Ginsberg, J. P. (2017). An Overview of Heart Rate Variability Metrics and Norms. Frontiers in Public Health. https://doi.org/10.3389/fpubh.2017.00258

[8] Google. MediaPipe Tasks Vision Package, privacy notice. https://github.com/google-ai-edge/mediapipe/blob/master/mediapipe/tasks/web/vision/README.md

[9] Bobbia, S., Macwan, R., Benezeth, Y., Mansouri, A., and Dubois, J. (2017). Unsupervised skin tissue segmentation for remote photoplethysmography. Pattern Recognition Letters. Dataset author page: https://sites.google.com/view/ybenezeth/ubfcrppg

Appendix A. Synthetic signal figures

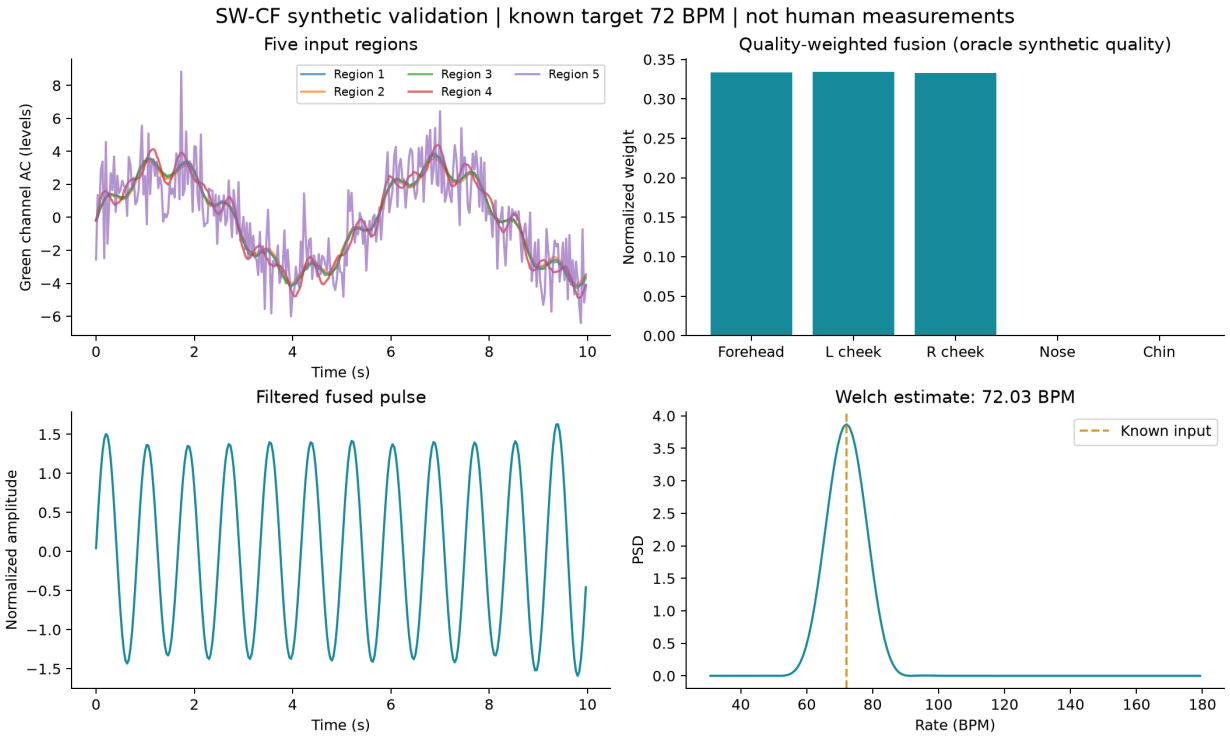


Figure A1. Artificial five-region RGB input, supplied-quality fusion weights, pulse waveform, and Welch PSD. The target is 72 BPM; this is not a human recording.

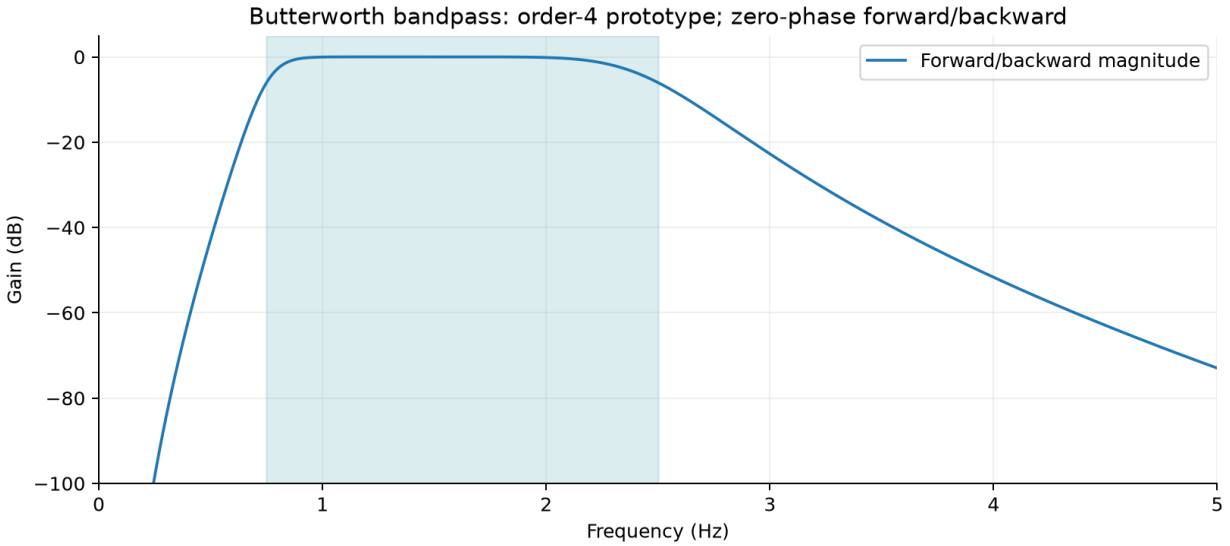


Figure A2. Combined forward/backward Butterworth magnitude response. Shading marks the chosen 0.75-2.5 Hz passband.

Appendix B. Demonstration dashboard

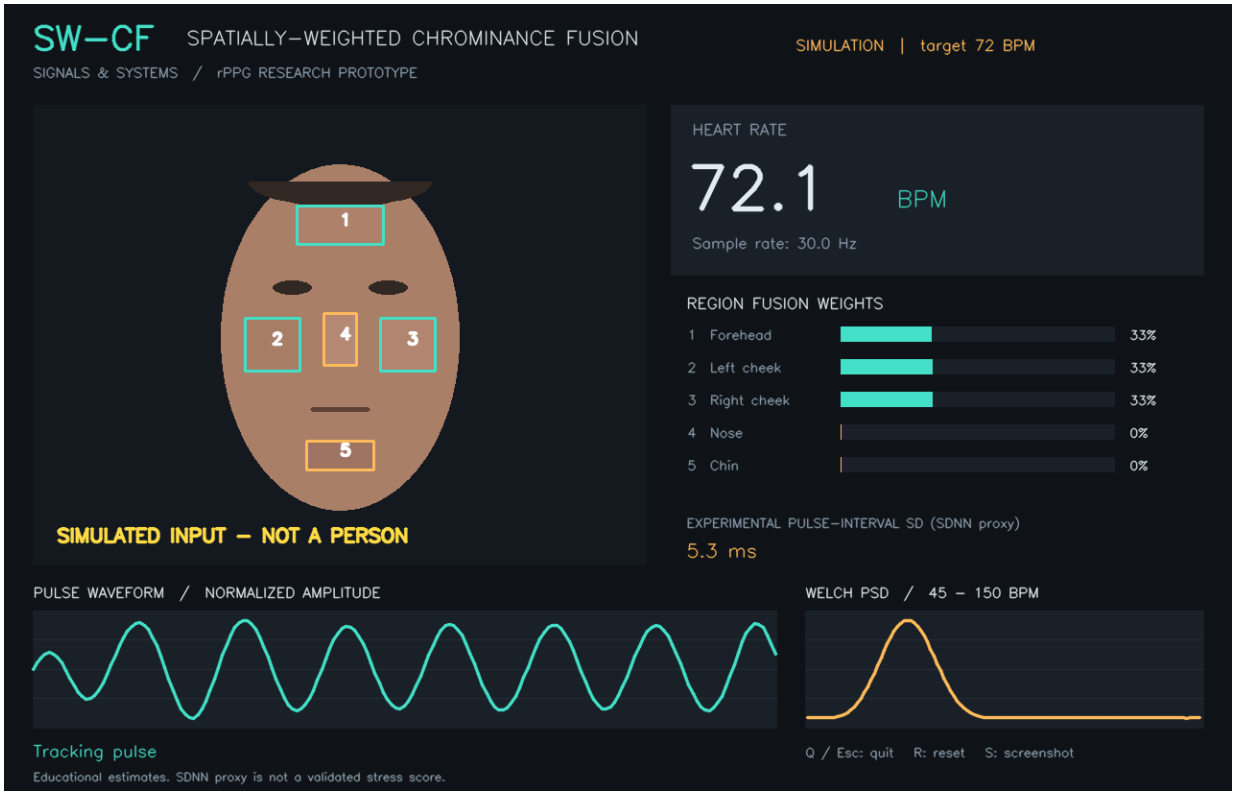


Figure B1. Saved synthetic dashboard after a 70-second generated signal. The target is 72 BPM. Two corrupted patches receive low supplied quality. All figures on this page are artificial; the experimental interval SD is processing/noise variation, not physiological variability.

Reproduce the demonstration

```
python main.py demo --headless --duration 70 --output outputs/demo
python scripts/validate.py
python -m pytest -q
```

Use the project's Python virtual environment. See README.md for setup and Windows commands. CSV evidence, scripts, test output, and the editable report source accompany this PDF.